

However, beyond B, the acceleration is positive and the velocity will get less negative and decreases in magnitude.

4.

Ans: **A**

$$F_{net} = ma$$

$$80 - 5(9.81)(\sin 30^\circ) - 4(9.81)(\sin 30^\circ) - 10 = (5.0 + 4.0)a$$

$$25.855 = 9.0a$$

$$a = 2.87 \text{ m s}^{-2}$$

$$F_{net} = ma$$

$$= 5.0(2.87)$$

$$= 14 \text{ N}$$

5.

Ans: **D**